

Progettazione di Applicazioni Mobili

Mobile app design

Evaluate

Stefano Burigat

Dipartimento di Scienze Matematiche, Informatiche e
Fisiche - Università di Udine

stefano.burigat@uniud.it



Evaluation: why

- To develop high quality apps it is necessary to test them with users as early and as often as possible
- Of key importance in this context is the user-interface, which builds the center of interaction between the application and its user
- With tests, design decisions can be evaluated and wrong decisions become apparent at a stage early enough to be still corrected in the further development

Evaluation: when

- Do not wait to have a complete prototype of the interface before getting feedback from potential users
- Doing evaluation late in the project does not fulfill the original promise of UX design, nor represent the optimal use of team's resources, particularly for mobile projects
- Build an initial version of the interface, and then test it with users and revise it as many times as you have money and time for

Evaluation: target users

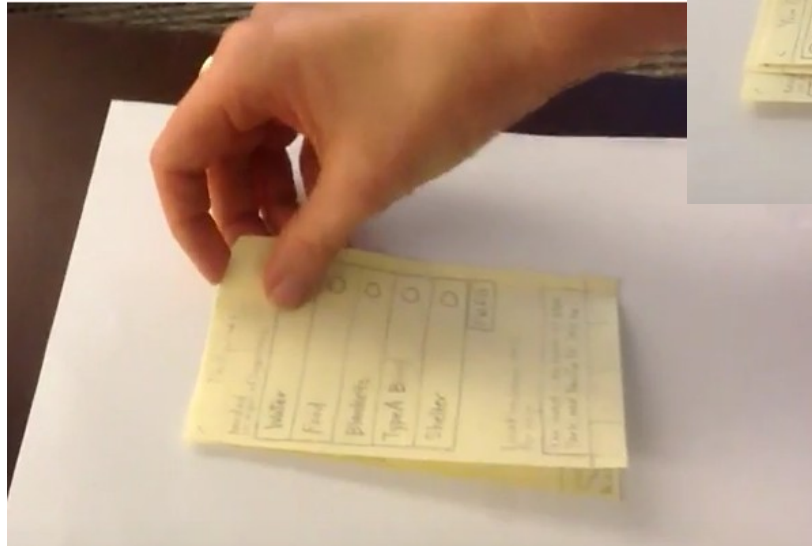
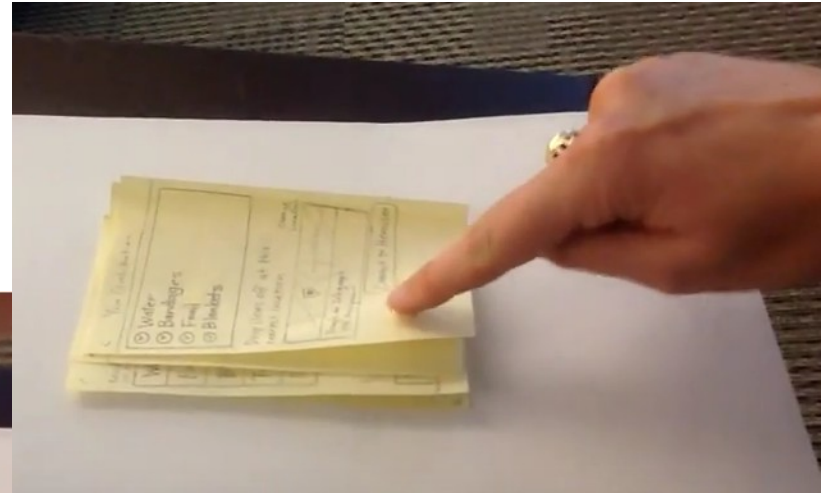
- The ideal scenario for evaluation is to find the exact representative of your target audience
- In practice, you can use anyone who is able, at some point in their life, to imagine asking their mobile device the questions your product or service is designed to answer
- Most apps don't require skills outside of the usual 80% of the mobile activities performed by a typical person living in any relatively wealthy country in the 21st century

Evaluation: documentation

- During the evaluation, capture what users are doing and saying, possibly using automated tools
- After you carry out the evaluation, you should document the results based on the information you collected
 - Document the results for each user first
 - Then group similar results to identify common issues
- You can use annotated task flows to clearly document each issue



Paper prototype evaluation



- Think about the tasks users should do with your app
- Explain each task to the users in as few words as possible and let them figure out how to do the task on their own using the interface (possibly using think aloud)
- Hand the user the starting screen for the task and add next screens on top (removing previous screens) when the user taps the appropriate interface element
- If users tap an interface element that wasn't part of your planned flow, ask them what they expected to see
 - you can even do a brief sketch of the missing screen in direct collaboration with the user

- Watch what they do first, then ask valuable questions (e.g., to clarify why they interacted with the wrong interface element to carry out a task)
 - Learn to ask non-leading questions
 - Answer questions with questions
 - Avoid calling interface elements with their technical names
- You can show and test transitions by using the appropriate motion when adding a screen

Digital prototype evaluation

- The rules for testing digital prototypes are the same as for paper prototypes
 - ... except that you can use a real mobile device and do not need to hand out screens manually
- For testing purposes, you can transform a paper prototype into a digital one by scanning your paper screens
 - ... then you can create an “interactive” prototype using tools such as Figma or Marvel
 - ... you just need to create transitions between screens when users interact with specific areas of the screen

- You can also include logging features in your digital prototypes to understand exactly what users do during tests
- Internal or external video-capture can also be used to understand what users are doing and/or what they are looking at on the screen



Five-second test

- In a five-second test, you expose users to a screen or moment in a product, ask them to look at it for five seconds, and then remove the screen from view
- Once the screen has been removed, ask users what they remember seeing, and what they thought the overall purpose of the page or screen was
- Considering that people often use products in a distracted, multitasking state, the five-second test is a pretty good indicator for how people really experience your products

- Did the user notice the most important messages or information that you're trying to convey in that moment?
 - If not, your information hierarchy may be off
- Did the user correctly interpret the purpose of the product and the screen?
 - If not, the balance of messaging and basic affordances (or, what it looks like you can do with that page) may need more work

- The five-second test is a great way to validate important parts of a product, like the start screen or the home page
- To test that the product design is robust at all levels, consider running five-second tests on random secondary screens as well
- Typically, the five-second test works better with medium-high fidelity prototypes

Black hat session

- A black hat session is a team activity
- In a black hat session, everyone has one job and one job only: to assume the most critical, judgmental perspective they can muster, and look at the designs from that point of view
- A black hat session can put your teammates in a state of mind that enables candid and productive feedback on designs

- A black hat session is valuable for several reasons:
 - It gives a group of people permission to call things as they see them, which can release them from any worry about whether their feedback is relevant or even correct
 - It gives anyone who is unhappy with certain aspects of the design a constructive forum for sharing his concerns and instills confidence that he is being heard
 - It creates a safe place where people who are less comfortable giving negative feedback can express their concerns as well
 - It puts all of these data points into a format that you can see, track, manage, and take charge of

Doing a Black Hat Session

- Explain the rules
- Give the team 15 to 20 minutes to
 - walk through all of the designs
 - reflect on what they notice
 - write sticky notes for every problem or issue they see
 - place the note on the designs near where they spotted the problem
- Review and look for themes and missing issues
- Engage the group in a discussion about the themes that emerged and record the points on a flipchart or white board

- Possible questions

- When you look at each screen or step, do you understand its fundamental purpose?
- What jumps out at you? Is it what should jump out at you?
- Do you know what you would click on or touch to advance to the next step?
- What questions do you have about the information and functionality that you're seeing?
- Are you satisfied that this is a reasonable number of steps?
- Is there anything that feels too complicated or cumbersome?
- Is there any language that doesn't make sense? Instructions? Labels on buttons? Anything else?